

Coding & Solution

Date	30 June 2026
Team ID	
Project Name	A comprehensive measure of well being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

- Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
- The solution should address the core problem statement and implement the key functional requirements.
- Include all source files, configuration files, and setup instructions needed to run the application.
- Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/PNT2026TMID18420/hdi-prediction-ml
Programming Language(s)	Python, JavaScript
Framework(s) Used	FastAPI, React, scikit-learn
Key Features Implemented	HDI score/tier prediction, Health/Education/Income sub-index breakdown visualization, CSV batch upload for multi-country comparison, PDF/CSV report export, role-based login (Admin/Analyst).
Pending / Incomplete Features	Automated live sync with the UNDP / World Bank APIs (currently supports manual CSV refresh).

Setup / Run Instructions	1) Clone the repository. 2) pip install -r requirements.txt. 3) Run the API with 'uvicorn app:app --reload'. 4) cd frontend && npm install && npm start. 5) Open http://localhost:3000 in a browser.
---------------------------------	--

Code Quality Checklist

S.No	Criteria	Status (Yes/No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

All core functional requirements are implemented and tested locally. The application runs cleanly with the sample UNDP/World Bank dataset; production deployment to AWS/Render is scheduled ahead of the final grading window.